

Unit-II

Transceiver Design Consideration in Physical Layer

The most crucial points influencing PHY design in wireless sensor networks are:

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- ❖ Low power consumption.
- ❖ As one consequence: small transmit power and thus a small transmission range.
- ❖ As a further consequence: low duty cycle. Most hardware should be switched off or operated in a low-power standby mode most of the time.
- ❖ Comparably low data rates, on the order of tens to hundreds kilobits per second, required.
- ❖ Low implementation complexity and costs.
- ❖ Low degree of mobility.
- ❖ A small form factor for the overall node.
- ❖ In general, in sensor networks, the challenge is to find modulation schemes and transceiver architectures that are simple, low-cost but still robust enough to provide the desired service.

- ❖ Energy usage profile
- ❖ Choice of modulation scheme
- ❖ Dynamic modulation scaling
- ❖ Antenna considerations

1. Energy usage profile:

- ❖ The choice of a small transmit power leads to an energy consumption profile different from other wireless devices like cell phones.
- ❖ The radiated energy is small, typically on the order of 0 dBm (corresponding to 1 mW). On the other hand, the overall transceiver (RF front end and baseband part) consumes much more energy than is actually radiated.
- ❖ Estimate that a transceiver working at frequencies beyond 1 GHz takes 10 to 100 mW of power to radiate 1 mW.
- ❖ Similar numbers are given for 2.4-GHz CMOS transceivers: : For a radiated power of 0 dBm, the transmitter uses actually 32 mW, whereas the receiver uses even more, 38 mW. For the Mica motes, 21 mW are consumed in transmit mode and 15 mW in receive mode.
- ❖ These numbers coincide well with the observation that many practical transmitter designs have efficiencies below 10 % at low radiated power.

1. Energy usage profile...

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- ❖ A second key observation is that for small transmit powers the transmit and receive modes consume more or less the same power; it is even possible that reception requires more power than transmission depending on the transceiver architecture, the idle mode's power consumption can be less or in the same range as the receive power.
- ❖ To reduce average power consumption in a low-traffic wireless sensor network, keeping the transceiver in idle mode all the time would consume significant amounts of energy.
- ❖ Therefore, it is important to put the transceiver into sleep state instead of just idling. It is also important to explicitly include the received power into energy dissipation models, since the traditional assumption that receive energy is negligible is no longer true.
- ❖ There is the problem of the startup energy/startup time, which a transceiver has to spend upon waking up from sleep mode,

1. Energy usage profile...

- ❖ A third key observation is the relative costs of communications versus computation in a sensor node. Clearly, a comparison of these costs depends for the communication part on the BER requirements, range, transceiver type, and so forth, and for the computation part on the processor type, the instruction mix, and so on.

2. Choice of modulation scheme:

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- ❖ A crucial point is the choice of modulation scheme. Several factors have to be balanced here: the required and desirable data rate and symbol rate, the implementation complexity, the relationship between radiated power and target BER, and the expected channel characteristics.
- ❖ To maximize the time a transceiver can spend in sleep mode, the transmit times should be minimized. The higher the data rate offered by a transceiver/modulation, the smaller the time needed to transmit a given amount of data and, consequently, the smaller the energy consumption.
- ❖ A second important observation is that the power consumption of a modulation scheme depends much more on the symbol rate than on the data rate.
- ❖ For example, power consumption measurements of an IEEE 802.11b Wireless Local Area Network (WLAN) card showed that the power consumption depends on the modulation scheme, with the faster Complementary Code Keying (CCK) modes consuming more energy than DBPSK and DQPSK

2. Choice of modulation scheme...

- ❖ m-ary modulation requires more complex digital and analog circuitry than 2-ary modulation, for example, to parallelize user bits into m-ary symbols.
- ❖ Many m-ary modulation schemes require for increasing 'm' an increased E_b/N_0 ratio and consequently an increased radiated power to achieve the same target BER; others become less and less bandwidth efficient.
- ❖ However, in wireless sensor network applications with only low to moderate bandwidth requirements, a loss in bandwidth efficiency can be more tolerable than an increased radiated power to compensate E_b/N_0 losses.
- ❖ It is expected that in many wireless sensor network applications most packets will be short, on the order of tens to hundreds of bits. For such packets, the startup time easily dominates overall energy consumption, rendering any efforts in reducing the transmission time by choosing m-ary modulation schemes irrelevant.
- ❖ The optimal decision would have to properly balance the modulation scheme and other measures to increase transmission robustness, since these also have energy costs:

2. Choice of modulation scheme...

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- ❖ With retransmissions, entire packets have to be transmitted again.
- ❖ With FEC coding, more bits have to be sent and there is additional energy consumption for coding and decoding. While coding energy can be neglected, and the receiver needs significant energy for the decoding process.
- ❖ This is especially cumbersome if the receiver is a power-constrained node.
- ❖ The cost of increasing the radiated power depends on the efficiency of the power amplifier, but the radiated power is often small compared to the overall power dissipated by the transceiver, and additionally this drives the PA into a more efficient regime.
- ❖ Specifically, the energy-per-bit consumption (defined as the overall energy consumption for transmitting a packet of n bits divided by n) of different m -ary QAM modulation schemes has been investigated for different packet sizes, taking startup energy and the energy costs of power amplifiers as well as PHY and MAC packet overheads explicitly into account.

3. Dynamic modulation scaling

- ❖ Even if it is possible to determine the optimal scheme for a given combination of BER target, range, packet sizes and so forth, such an optimum is only valid for short time;
- ❖ as soon as one of the constraints changes, the optimum can change, too. In addition, other constraints like delay or the desire to achieve high throughput can dictate to choose higher modulation schemes.
- ❖ Therefore, it is interesting to consider methods to adapt the modulation scheme to the current situation. Such an approach, called **dynamic modulation scaling**,
- ❖ In particular, for the case of **m-ary** QAM and a target BER of 10^{-5} , a model has been developed that uses the symbol rate B and the number of levels per symbol m as parameters.
- ❖ This model expresses the energy required per bit and also the achieved delay per bit (the inverse of the data rate), taking into account that higher modulation levels need higher radiated energy.
- ❖ The energy per bit depends much more on m than on B . In fact, for the particular parameters chosen, it is shown that both energy per bit and delay per bit are minimized for the maximum symbol rate.

3. Dynamic modulation scaling...

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- ❖ The modulation scaling, a packet is equipped with a delay constraint, from which directly a minimal required data rate can be derived.
- ❖ Since the symbol rate is kept fixed, the approach is to choose the smallest m that satisfies the required data rate and which thus minimizes the required energy per bit. Such delay constraints can be assigned either explicitly or implicitly.
- ❖ When there are no packets present, a small value for m can be used, having low energy consumption. As backlog increases, m is increased as well to reduce the backlog quickly and switch back to lower values of m . This modulation scaling approach has some similarities to the concept of **dynamic voltage scaling**.

4. Antenna Consideration

- ❖ In small form factor of the overall sensor nodes restricts the size and the number of antennas. As explained above, if the antenna is much smaller than the carrier's wavelength,
- ❖ it is hard to achieve good antenna efficiency, that is, with ill-sized antennas one must spend more transmit energy to obtain the same radiated energy.
- ❖ Secondly, with small sensor node cases, it will be hard to place two antennas with suitable distance to achieve receive diversity.
- ❖ The antennas should be spaced apart at least 40–50 % of the wavelength used to achieve good effects from diversity. For 2.4 GHz, this corresponds to a spacing of between 5 and 6 cm between the antennas, which is hard to achieve with smaller cases.
- ❖ The radio waves emitted from an antenna close to the ground – typical in some applications – are faced with higher path-loss coefficients than the common value $\propto 2$ for free-space communication.
- ❖ Typical attenuation values in such environments, which are also normally characterized by obstacles.

4. Antenna Consideration

- ❖ Depending on the application, antennas must not protrude from the casing of a node, to avoid possible damage to it. These restrictions, in general, limit the achievable quality and characteristics of an antenna for wireless sensor nodes.
- ❖ Nodes randomly scattered on the ground, for example, deployed from an aircraft, will land in random orientations, with the antennas facing the ground or being otherwise obstructed.
- ❖ This can lead to non isotropic propagation of the radio wave, with considerable differences in the strength of the emitted signal in different directions.
- ❖ This effect can also be caused by the design of an antenna, which often results in considerable differences in the spatial propagation characteristics (so-called lobes of an antenna)